Within- and between-year patterns of Seasonal Influenza and Respiratory Syncytial Virus and the potential for concurrent outbreaks

Keywords: respiratory pathogens, influenza, RSV, seasonal dynamics, survival analysis, concurrent outbreaks, geographic variation

1. Introduction (644 words)

Influenza viruses and respiratory syncytial virus (RSV) are respiratory pathogens that impose a large disease burden worldwide, causing millions of infections and substantial morbidity and mortality, including hundreds of thousands of deaths each year. In many temperate and some subtropical regions, the disease burden associated with these pathogens exhibits strong within-year cyclical patterns, often peaking in winter or cooler months. Such within-year patterns may be driven by environmental and climatic factors (e.g. temperature and humidity), human behaviours (e.g. school terms, holiday travel, and indoor crowding), and immune dynamics (e.g. waning or boosting of population immunity). These within-year patterns vary by region and pathogen. Between-year cyclical patterns may also occur, driven by changes in circulating pathogens (e.g., antigenic drift, antagonistic or synergistic interactions), demographics (e.g., population turnover), environmental and climatic patterns (e.g., El Niño/La Niña), immune dynamics (e.g., waning immunity), and behaviours (e.g., more risk-averse behaviors following large outbreaks).

Where within- and between-year patterns play a role, the circulation of seasonal respiratory pathogens may peak concurrently. Such concurrent outbreaks of seasonal respiratory pathogens have been observed in many health systems, such as in Europe, the United States, and China, most of which involve both the influenza virus and RSV. These concurrent outbreaks pose substantial challenges to public health systems, as they can lead to increased demand for health services, and can consequently disrupt the timeliness of vaccination and other preventive measures. Therefore, to better support health systems' preparation for "winter pressures", it is crucial to understand the risk of concurrent respiratory disease outbreaks. In this study, we address this knowledge gap by quantifying the probability and

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interval to the next concurrent outbreak of seasonal influenza and RSV.

By far, most relevant research has focused on deriving and understanding the within-year patterns of seasonal respiratory pathogens. The between-year patterns, however, have rarely been considered due to data constraints. Such analysis requires long-term surveillance data at fine resolution to be simultaneously available for multiple pathogens. Coarse data resolution, such as provincial level aggregate data, often masks spatial variations in both within- and between-year patterns, and thus dampens out the potential signal for concurrent outbreaks. Furthermore, validating the role played by between-year patterns requires multiple datasets, which in turn requires the type of data mentioned above—already rare—to be available for multiple locations. In this study, we put together a dataset describing the burden of seasonal influenza and RSV for eight cities in mainland China to address these key data challenges.

Further, most respiratory disease prediction models are designed to forecast for the numeric disease burden of interest attributable to single pathogens (e.g. x number of RSV hospitalisations are expected in the next y weeks). Here, we provide a complementary framework to provide seasonal level forecasts for the probability of a concurrent outbreak and the possible temporal interval until the subsequent concurrent outbreak, covering a different aspect of respiratory public health risks important to the health system. Seasonal level forecasts have been widely used in the context of vector-borne infectious diseases, and therefore their value to managing respiratory health warrants further exploration. To achieve this, we face two key technical challenges. First, methods designed for identifying single cyclical components may mischaracterise between-year patterns as residual noises. Second, for most locations, concurrent outbreaks are rare events. With finite observational window, it is challenging to distinguish between "will not occur" and "have not yet occurred" events.

Using long-term surveillance data of seasonal influenza and RSV longer than five years for eight cities in mainland China spanning various climate zones, we developed an analytical framework that combines time-series statistics and survival analysis to quantify the probability of a concurrent outbreak and the interval to the subsequent concurrent outbreak. Although the estimates we provide focus on seasonal influenza and RSV, our framework is widely adaptable to studying other respiratory pathogens with continuous measured data describing their disease burden.

2. Methods (1283 words)

2.1. Data Sources and Data Preprocessing

We collected long-term surveillance data (>= five years) of seasonal influenza and RSV at the city level in mainland China with a temporal resolution of month or finer from published literature. Since it is a requirement for our analysis that influenza and RSV data be simultaneously available for a given city, and that the historical surveillance infrastructure has been more complete for influenza than RSV in the context of mainland China, we started out to identify data sources on RSV disease burden first. A recent study by Guo et al. (2024) provided a comprehensive review of the literature on RSV disease burden in mainland China. Among the studies they identified, we found X datasets to meet our criteria of (1) >= five years of data; (2) monthly or finer temporal resolution; (3) city level or finer geographic resolution. These datasets were then supplemented with a broader search of the literature through PubMed. Search terms we used are included in the Supplementary Material Table S1.

This process has helped us identify eight cities to focus on for our search for influenza burden data: Beijing (provincial level city), Lanzhou (Gansu province), Xi'an (Shaanxi province), Wuhan (Hubei province), Suzhou (Jiangsu province), Wenzhou (Zhejiang province), Guangzhou (Guangdong province), and Yunfu (Guangdong province). Of these cities, Beijing, Lanzhou, and Xi'an are located in the northern part of mainland China, and the rest are located in the southern part. We then searched for influenza burden data in these same eight cities using the same set of search terms in PubMed, replacing the search term "RSV" with "Influenza"; the search term "China" with the specific city names. Data embedded in figures were digitised using $XXXX$.

The specific metric describing the disease burden of influenza and RSV varied by city and by pathogen (Figure 1). However, their relative changes provided the key information describing the within- and between-year patterns. There was a mix of in- and out-patient settings. Of all city and pathogen combinations, seasonal influenza in Xi'an is the only situation where diagnostics were syndromic only. All other disease burden metrics were confirmed by laboratory diagnostics. More details of the disease burden metrics are presented in the Supplementary Methods Section 1.

Due to the non-negative nature of disease burden metrics, we transformed raw values by taking the natural logarithm before further analysis. Zeros were replaced with a small constant. All analyses were conducted on the datasets' original time scale (e.g. with weekly data, we tested if within-year cycle length was approximately 52 weeks). Results were converted to a monthly scale only for comparison and interpretation purposes.

2.2. Identifying within- and between-year cycles

We used a smoothing-based algorithm to derive the within- and between-year cycles in the disease burden of seasonal influenza and RSV. The foundation of this algorithm is LOESS (Locally Estimated Scatterplot Smoothing). Seasonal-Trend decomposition using LOESS (STL) is commonly used to derive within-year cycles in time-series data, assuming single

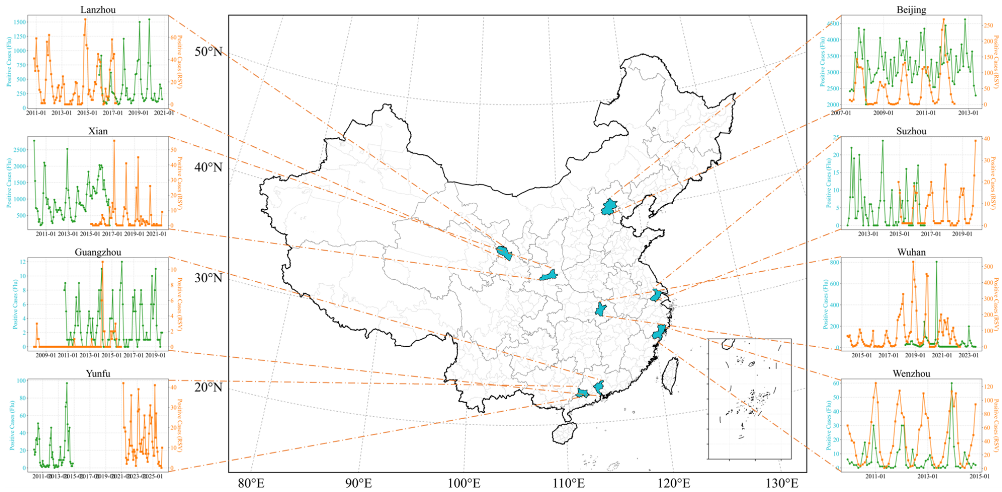

Figure 1: Selection and compilation of influenza and RSV data across the eight study cities. (a) Data selection workflow; (b) Geographic coverage and data representation.

internal cycles. The smoothing-based approach is numerically more flexible compared to other approaches that require researchers to make assumptions about expected mathematical forms (e.g. cosine models or cubic spline-based models) [**?**].

We further used MSTL (Multiple Seasonal-Trend decomposition using LOESS), which allows for more than one internal cycle. With MSTL, the original time-series were decomposed into four components: within-year patterns, between-year patterns, long-term trends, and residuals. We assumed the within-year cycle lengths to be between 10 and 14 months and between-year cycle lengths to be between 18 and 36 months. The upper limit of the between-year cycle lengths was further constrained by the total lengths of the series. For time series that do not contain at least two full cycles of a given period length, MSTL does not test that between-year cycle length, as this is a requirement of the MSTL algorithm. Model selection was performed using Akaike Information Criterion (AIC). The optimal model for each city-pathogen combination was defined as having the minimum AIC value. Following standard practice, we considered models with $\Delta\text{AIC} < 7$ (where $\Delta\text{AIC} = \text{AIC}_i - \text{AIC}_{\text{min}}$) as having comparable empirical support. The optimal models and their comparable models were all included in further analysis (i.e. forward simulation and concurrent outbreak identification). The lowest AIC estimates achievable using a single-cycle model were compared against the lowest AIC estimates achievable using multi-cycle models. If the latter is lower, then we conclude that between-year patterns play a role in predicting the burden in the context of the city-pathogen combination. Bayesian Information Criterion (BIC) was also examined as a sensitivity analysis.

2.3. Forward simulations of outbreaks and concurrent outbreak identification

To assess the implications of within- and between-year patterns on future disease burden, we performed forward simulations based on observed disease burden history and the optimal and their comparable models derived using the methods above. These forward simulations were performed for long timeframes to ensure that within- and between-year patterns estimated using different time-series may be overlaid for a given city. Within- and between-year patterns were simply repeated in these forward simulations. Trend and residual components of the optimal models were fitted to an Auto-Regressive Integrated Moving Average (ARIMA) model. This ARIMA model was then used to generate 1000 stochastic forward trajectories of the trend and residual components per model selected. Since the ARIMA model was fitted to a seasonally adjusted time series, the mean-reverting properties ensure that long-term forward simulations remain bounded. This stochastic simulation framework was developed to characterise the uncertainty around event probabilities, rather than providing forecasts of future disease activities. When a given city-pathogen combination has n optimal models, then $1000 \times n$ stochastic forward trajectories were generated. All forward simulations were performed based on their original time scale, as presented in the source literature.

Using the forward simulations of both seasonal influenza and RSV disease burden, we explore the possibility of concurrent outbreaks across different outbreak thresholds. The measure of seasonal influenza and RSV disease burden relied on different epidemiological measures depending on the city and health systems (e.g. attributable hospital admissions, test positive rates) and therefore cannot be directly compared on their original scale. Therefore, we rescaled these measures by decile, leaving the 1st and 10th deciles unbounded. Concurrent outbreaks were defined as when seasonal influenza and RSV disease burden simultaneously exceeded the 70th, 80th, and 90th percentiles of their observed disease burden history. A year is considered a concurrent outbreak year if the concurrent outbreak threshold has been exceeded at least once in that year.

2.4. Characterising concurrent outbreak frequency using survival analysis

Inter-concurrent outbreak intervals were calculated as the time elapsed between successive concurrent outbreak years. Directly counting the intervals between concurrent outbreaks over-weights short intervals as they have more placement opportunities given the finite forward simulation window. It also introduces right-censoring issues when no concurrent outbreak or only one concurrent outbreak was observed in the forward simulation window. To address these issues, we reframed the interval detection as a time-to-event problem, and estimated the inter-concurrent outbreak intervals using Kaplan-Meier survival analysis, with left truncation and right-censoring. The survival curve was fitted for each city at all three different concurrent outbreak thresholds defined above. With this setup, we can calculate inter-concurrent outbreak intervals. In the context of this study, we used the time by which 50% of the forward simulation paths ($1000 \times n$ models) are expected to experience their subsequent concurrent outbreak event as the inter-concurrent outbreak interval.

2.5. Software and code availability

All aforementioned methods and analyses were implemented in R (v4.2.2) using the following `packages`: `forecast`, `mstl`, and `survival`. All codes are available on the GitHub repository: https://github.com/yangclaraliu/respiratory_multi_pathogen_seasonality.

3. Results (1042 words)

3.1. Model comparison

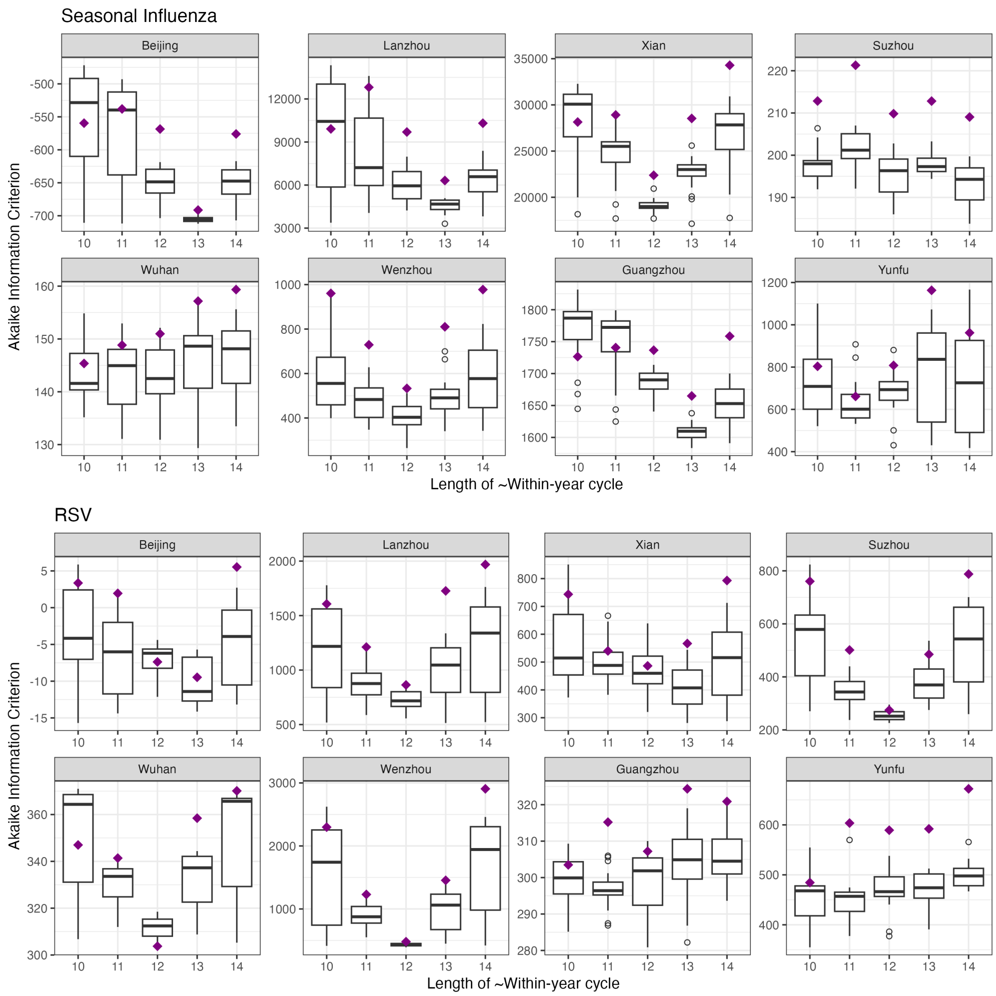

Figure 2: The association between within-year cycle lengths and Akaike Information Criterion (AIC) values by pathogen and city. Each purple point represents a model with only within-year cycles. Each boxplot represents the distribution of AIC values for models with the corresponding within-year cycle length and the full range of between-year cycle lengths tested.

We compared the performance of the models only adjusted for within-year cycles against the models adjusted for both within- and between-year cycles using AIC values (Figure 2). AIC values combine model fit (log-likelihood) with complexity penalties, and therefore the scales depend on both data characteristics and model structure. Therefore, AIC value comparisons are only meaningful if the underlying data set remains the same (i.e. the same city and pathogen combination). Lower AIC values indicate better model performance. In nearly all combinations of pathogen, city, and within-year cycle length, some models adjusted for both within- and between-year cycles performed better than the models only adjusted for within-year cycles after penalising for increased degrees of freedom. The only exceptions were RSV in Wuhan - models with only annual cycles performed better than all other models tested. Thus, for seasonal influenza and RSV, significant between-year cycles may improve model performance beyond annual cycles alone.

For RSV, the relationship between within-year cycle length and AIC was often U-shaped, with the minimum at 12 months, indicating a best-fitting annual cycle across many city–pathogen combinations. In contrast, seasonal influenza showed a less consistent pattern: the AIC minimum frequently deviated from 12 months, implying that the optimal seasonal period is not strictly annual. This could reflect time-varying or heterogeneous periodicity (e.g., episodes of slightly shorter or longer cycles that average to ≈ 12 months over the long term) or stochastic variation, despite long-term epidemiology appearing annual.

In several panels, the AIC values near the minimum were very densely distributed (i.e. small dispersion), indicating that multiple model specifications produce comparably performing models. Accordingly, in forward simulations and concurrent outbreak analysis, we carried forward both the optimal models and their comparable models (ΔAIC < 7). The number of comparable models for influenza ranged from 2 (in Xi'an, Lanzhou, Wenzhou, and Yunfu) to 20 (in Wuhan and Beijing); that for RSV ranged from 2 (in Wenzhou and Yunfu) to 39 in Beijing. The full list detailing the number of comparable models by city and pathogen is available in the Supplementary Material (Table S1).

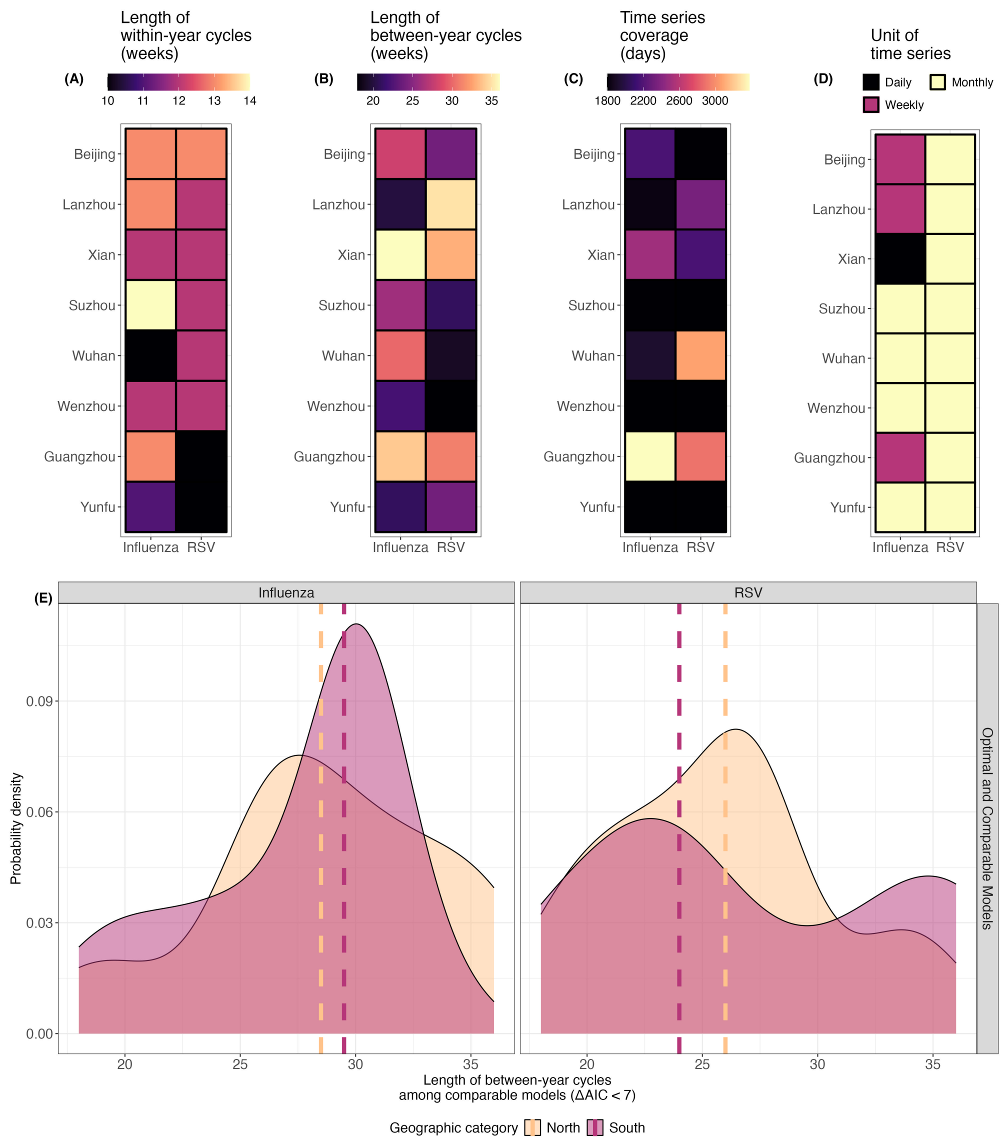

Figure 3: Seasonal component analysis and geographic patterns.

189 The optimal within-year cycle lengths and between-year cycle lengths for each city and pathogen combination are presented in Figure 3 (A) and (B). These results reflected only the optimal models—their comparable models are not included. Similar to the results presented in Figure 2, most within-year cycles were approximately 12 months for RSV and were not strictly annual for seasonal influenza. The between-year cycles selected for the optimal models were much more variable. These results are not strongly associated with the length of the time series (Figure 3(C)) or their temporal resolution (Figure 3(D)). Looking across all optimal and their comparable models, the between-year cycles for influenza had a mean of 28 months in the North and 27 months in the South; those for RSV had a median of 25 months in the North and 28 months in the South (Figure 3(E)). These differences are not

statistically significant by region for either pathogen, likely because of small sample size and thus limited statistical power. The between-year cycles were highly variable (i.e. IQR > 6 months) for Suzhou (influenza), Wuhan (RSV) and Guangzhou (RSV) (see Supplementary Material Figure S1).

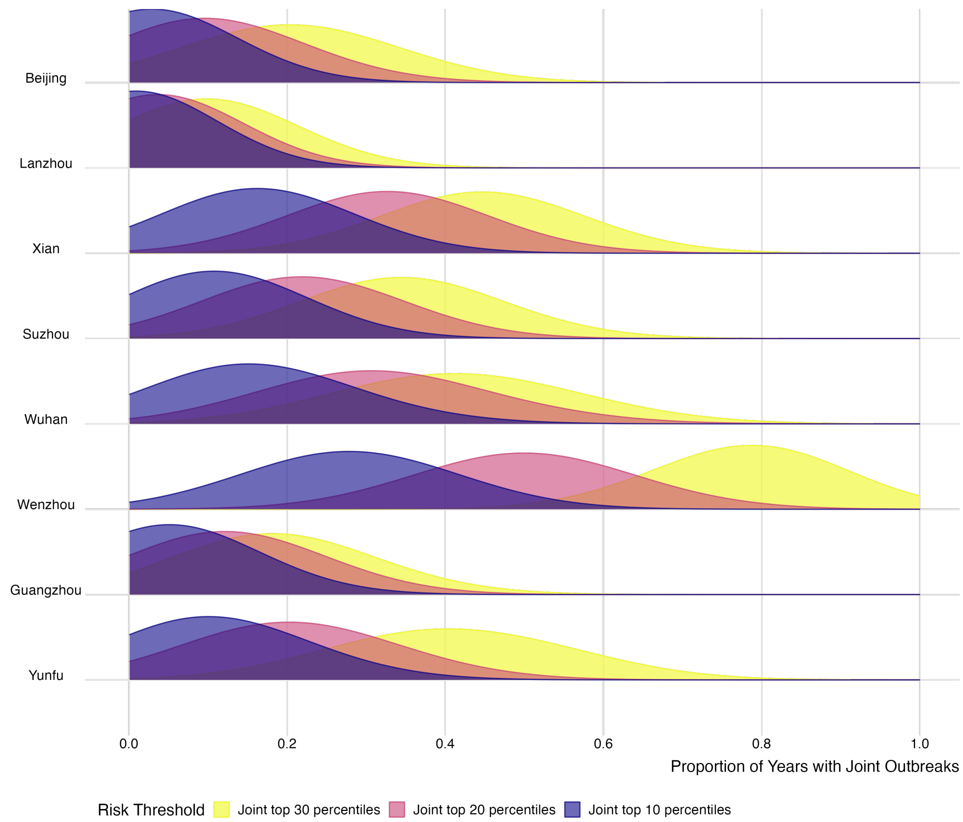

Figure 4: Proportion of years having concurrent outbreaks of seasonal influenza and RSV under different risk thresholds by city.

Forward projection not only reflects the within- and between-year cycles we estimated, but also the underlying city-specific epidemic histories. Using both optimal and comparable models, we demonstrate that there are non-negligible risks of concurrent outbreaks of influenza and RSV in the cities we investigated (Figure 4). Using the strictest risk threshold (joint top 10 percentile, i.e. simultaneously observing influenza and RSV exceeding the 90th percentile of the city's respective observed history)), Wenzhou was projected to experience the most frequent concurrent outbreaks (i.e. 25% of years projected), followed by Xi'an and Wuhan (15% of years projected). Beijing and Lanzhou were projected to not experience any concurrent outbreaks given this strict risk threshold. Under less strict thresholds, we found that the ranking by frequency remains consistent. When concurrent outbreak was defined by joint top 20 percentile (i.e. simultaneously observing influenza and RSV exceeding the

80th percentile of the city's respective observed history), Beijing and Lanzhou were expected to observe some concurrent outbreaks (i.e. 10% and 5.9% of years projected, respectively). A complete table of median values of distributions presented in Figure 4 can be found in the Supplementary Material Table S2.

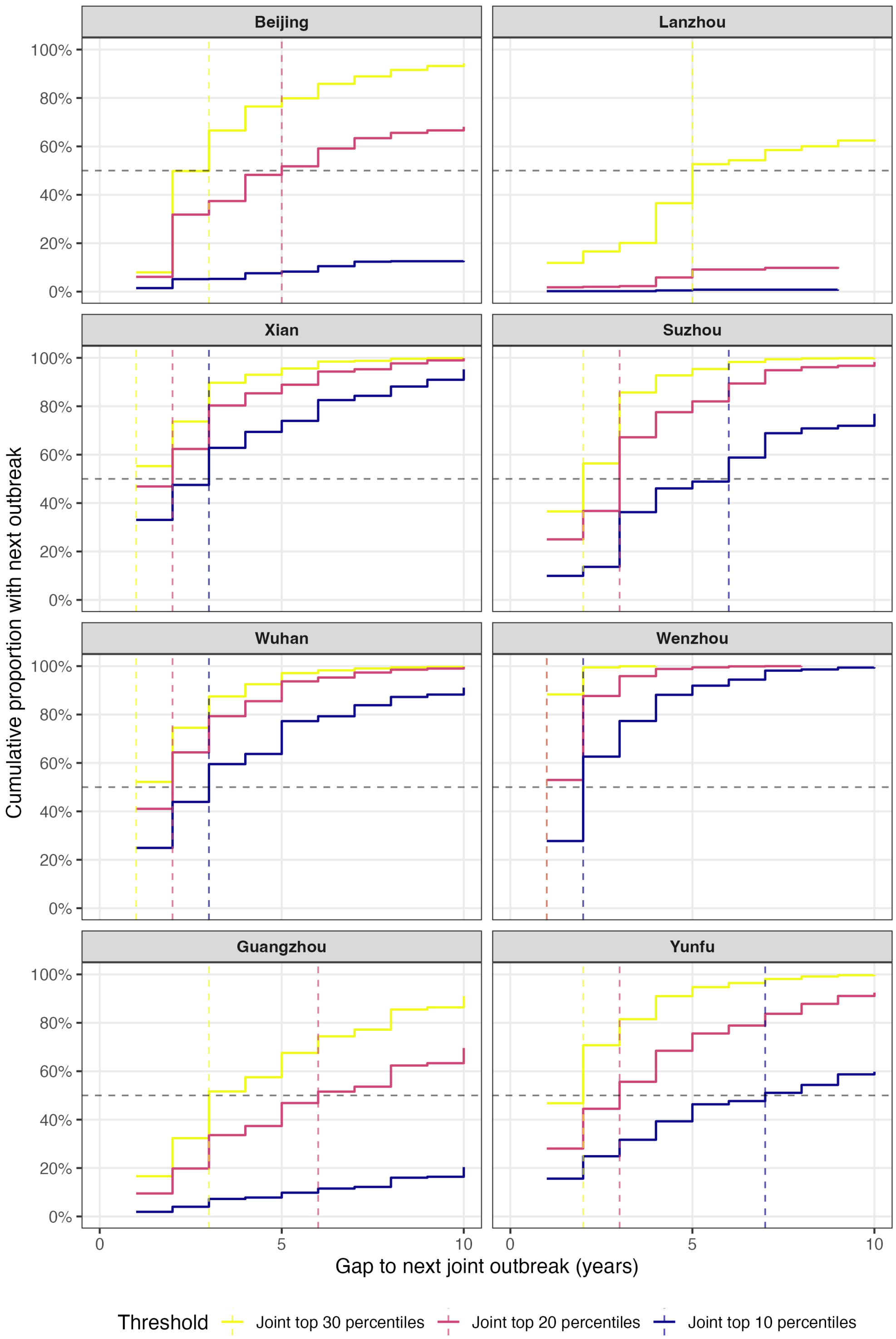

Figure 5: Interval to the subsequent concurrent outbreak of seasonal influenza and RSV under different risk thresholds by city.

218 Using the observed intervals between consecutive concurrent outbreak years from the
219 forward simulations, we estimated the distribution of temporal gaps to the next potential

concurrent outbreak for each city and threshold combination (Figure 6) using Kaplan-Meier survival analysis. Under the strictest risk threshold (joint top 10 percentile), the median temporal gap to the next concurrent outbreak was the shortest in Wenzhou, Xi'an, and Wuhan (two, three, and three years respectively). In other words, in these cities, at least 50% of the projection paths experienced another concurrent outbreak within two or three years after the last concurrent outbreak. In Suzhou and Yunfu, we expected longer temporal gaps between concurrent outbreaks (six and seven years, respectively). In Beijing, Lanzhou, and Guangzhou, concurrent outbreaks as defined by the strictest threshold were not likely based on the projection paths. Relaxing the joint epidemic threshold to the top 20 percentile (i.e. simultaneously observing influenza and RSV exceeding the 80th percentile of the city's respective observed history), the median temporal gap ranged from one to six years. Beijing and Guangzhou, which previously were not expecting any concurrent outbreaks, were now projected to experience them next in five or six years, respectively. Lanzhou was still projected to not experience any concurrent outbreaks given this outbreak risk threshold. A full table of the results can be found in the Supplementary Material Table S3.

4. Discussion (1491 words)

We used publicly available surveillance data on two major seasonal respiratory pathogens, seasonal influenza and respiratory syncytial virus (RSV), with at least 5 years of history from eight cities in mainland China to investigate within- and between-year patterns of their disease burden. Time-series analysis revealed that strong within-year cycles are present for both pathogens, consistent with the established understanding that they are fundamentally seasonal respiratory infections. Our model selection process further revealed that introducing between-year cycles can improve model fit, as measured by AIC, for both pathogens across nearly all cities (with the exception of RSV in Wuhan). We then combined time-series and survival analysis to quantify the probability of concurrent influenza–RSV outbreaks and the temporal gaps until the subsequent event of such nature under multiple risk thresholds. This analysis accounted for uncertainty around our models and the intrinsic stochasticity of the underlying data generation processes. The durations of between-year cycles, probabilities of concurrent outbreaks, and the temporal gaps until the subsequent concurrent outbreak all varied by city with no apparent geographic patterns.

To our best knowledge, our study is the first to quantify the probabilities of concurrent outbreaks of seasonal respiratory pathogens and the temporal gaps until the subsequent concurrent outbreak under multiple risk thresholds. Most existing literature tends to focus on quantifying annual cycles. Only a small number of studies have quantified non-annual cycles [**?** **?**] or confirmed the possibility of concurrent outbreaks of multiple seasonal respiratory pathogens [? ?]. Therefore, a key point of novelty of our study is methodological. The survival analysis approach we incorporate into the predominant time-series-based theoretical framework allowed us to address two key technical challenges in outbreak prediction: finite sample bias and right censoring. An example of finite sample bias is that over a 10-year period, concurrent outbreaks one year apart may occur nine times, whereas concurrent outbreaks six years apart may only occur once. In the distribution of raw temporal gaps, a strong bias towards shorter gaps would occur. Right censoring occurs when we have to draw an end for the analytical window although the event of interest may not have happened yet. Without addressing this issue, "not yet happened" may be misinterpreted as "would not happen". The relevance of this approach extends beyond influenza and RSV and can be broadly applicable to other respiratory diseases with continuously measured burden metrics.

In this study, concurrent outbreak risks were defined as the probability of simultaneously reaching the 70th, 80th, and 90th percentiles of historical values. By evaluating multiple thresholds, our framework accommodates different risk tolerance levels among decision makers—from those preparing for only the most severe concurrent outbreaks (90th percentile) to those planning for more frequent but less extreme events (70th percentile)—and quantifies how these choices affect estimated probabilities and recurrence intervals. This design enabled us to compare disease burden measures across cities and pathogens on a relative scale. This relative-scale design also reflects health system context—high health system pressure in Beijing and Yunfu would not look the same and cities should be evaluated relative to their own historical baseline.

We used a unique approach to time-series projection while assessing the probability of concurrent outbreaks and the temporal gaps to the subsequent concurrent outbreak, through which we propagate uncertainty from the model selection and data generation processes. The increase in variance further into the future is suppressed because such large noise would cloud the signal in the long run. Therefore, the results of our study can be interpreted as an assessment of rare events in the future given our current understanding of the underlying disease dynamics, and should not be interpreted as exact forecasting of concurrent outbreaks occurring in a specific year. Further, concurrent outbreak risks were assessed on the seasonal level. This type of prediction is similar to those of some climate-sensitive infectious diseases that have been used in public health decision-making. Such a seasonal lens should complement real-time tools forecasting for the exact number of public health endpoints in the context of respiratory public health management.

We found that RSV exhibited more strictly annual within-year cycles (i.e., exactly 12 months) than seasonal influenza, which is consistent with our existing understanding of these pathogens' epidemiology. RSV shows a stronger seasonal signal because its transmission relies more heavily on susceptibles from young age groups and experiences less intense competition between subtypes. While between-year cycles for RSV have been observed elsewhere in the world, those of seasonal influenza have not been widely studied. Here, we estimated highly variable between-year cycles by city and by pathogen, indicating that the drivers of these cycles may be complex and context-specific, potentially composed of multiple slowly varying factors such as antigenic drift, viral interference, demographic turnover, behavioural changes, and immunisation history. Notably, the durations of within- and between-year cycles are not directly deterministic of the probability of concurrent outbreaks or the temporal gaps to subsequent concurrent outbreaks. This indicates that predicting concurrent outbreaks requires consideration of not only pathogen characteristics and epidemiology but also the underlying transmission history.

Seasonal-level predictions of concurrent respiratory disease outbreaks are valuable for public health practitioners making early-season intervention decisions and for hospital system administrators seeking to prevent overcrowding and excessive strain on healthcare staff. Based on our estimates, communities in Xi'an, Wuhan, and Wenzhou may experience frequent concurrent outbreaks (i.e. temporal gaps to the subsequent concurrent outbreak are 2-3 years based on the strictest risk threshold) compared to other cities we investigated, and their local health systems may be better adapted to handling them. Our results provide evidence for the potential of further disease burden reduction if awareness of concurrent outbreaks is raised and early intervention strategies are implemented. In cities like Suzhou and Yunfu, however, concurrent outbreaks are less frequent (i.e. temporal gaps to the subsequent concurrent outbreak are 6-7 years based on the strictest risk threshold). At lower frequencies, a concurrent outbreak may affect the community more severely, as baseline health systems are less prepared for such events. In these cities, seasonal-level concurrent outbreak predictions may be particularly useful when integrated with other real-time forecasting tools. Decisions may need to balance the trade-off between lower frequency and potentially larger negative impact.

There are, however, limitations to our study. First, we limited our analysis to influenza and RSV data from eight cities in mainland China. Our approach requires high-resolution surveillance data (i.e., city level) with long histories due to the nature of our analytical framework, which constrains the number of locations where this analysis can currently be applied. As new testing products that cover more pathogens become available, our approach will be applicable in more locations. While the approach we propose is broadly generalisable, the specific results are not. The cities we investigated are located in subtropical and temperate climates, where within-year cycles are strong for both influenza and RSV. This would not be the case in tropical regions, where seasonal patterns may be less pronounced or absent. Second, we used only one measure of disease burden per city and per pathogen. In practice, multiple measures of disease burden may be available (e.g., hospitalisations, deaths, outpatient visits, test positivity rates, etc.). Pooling all available information could improve our understanding of the underlying cycles and enhance prediction accuracy, but such harmonization would require careful consideration of the underlying surveillance systems and their biases. Third, while we tested the predictive power of within- and between-year cycles, the underlying mechanisms driving these cycles warrant further investigation. Finally, the longest between-year cycle we tested was 36 months, constrained by the available time-series data. Longer-term data made possible by electronic health systems may allow us to test for even longer cycles in the future.

Future research should explore several directions to enhance the practical utility of seasonal-level concurrent outbreak predictions. First, integration with real-time forecasting tools would provide more comprehensive situational awareness for public health decision-making. This integration would require incorporating real-time data streams, which introduce technical challenges such as backfilling and data quality control. Since concurrent outbreak risks are essentially a time-to-event problem, real-time systems would need to dynamically update predictions based on recent transmission history. For example, predictions should be adjusted when no concurrent outbreaks have occurred for an extended period (e.g. two years), reflecting the changing conditional probability as time since the last event increases. Second, prospective validation of our forward simulation approach against observed concurrent outbreaks would strengthen confidence in these predictions and help distinguish between the probabilistic assessments we provide and true point forecasts. Third, incorporating additional data sources could improve prediction accuracy. These might include information about antagonistic or synergistic relationships between circulating pathogens (such as viral interference effects), changes in population immunity from vaccination campaigns, and shifts in pathogen antigenic characteristics that may influence the probability and timing of concurrent activity. Fourth, extending this framework to other combinations of co-circulating respiratory pathogens beyond influenza and RSV would test the generalisability of the approach and potentially reveal whether predictable patterns of concurrent circulation exist across broader pathogen communities.